

Question 1

Tuesday, 12 May 2026 10:38

We have the following transition and emission matrices:

Transition

From\To	H	L
H	0.5	0.5
L	0.4	0.6

Emission:

	State H prob	State L prob
A	0.2	0.3
C	0.3	0.2
G	0.3	0.2
T	0.2	0.3

Now Given the sequence: $S = ACCGTGCA$

We use Viterbi:

Denoting s_i – state i , v_i – probability of i th step

$s_0 = H$

$$\delta_1(H) = t(H|H)e(A|H) = 0.5 * 0.2 = 0.1$$

$$\delta_1(L) = t(L|H)e(A|L) = 0.5 * 0.3 = 0.15$$

Filling out the full table, with pointers to pervious cell:

$\delta \backslash i$	1(A)	2(C)	3(C)	4(G)
$\delta(H), prev is$	0.1	$0.1 * 0.5 * 0.3 = 0.015$	$0.018 * 0.5 * 0.3 = 0.0027$	$0.0027 * 0.5 * 0.3 = 4.05 * 10^{-4}$
$\delta(H), prev is$	-	$0.15 * 0.4 * 0.3 = 0.018$	$0.18 * 0.4 * 0.2 = 0.00144$	$0.0027 * 0.4 * 0.2 = 2.592 * 10^{-4}$
$\delta(L), prev is$	0.15	$0.1 * 0.5 * 0.2 = 0.01$	$0.018 * 0.5 * 0.2 = 0.0018$	$0.0027 * 0.5 * 0.2 = 2.7 * 10^{-5}$
$\delta(L), prev is$	-	$0.15 * 0.6 * 0.2 = 0.018$	$0.6 * 0.018 * 0.2 = 0.00216$	$0.00216 * 0.6 * 0.2 = 2.592 * 10^{-4}$

$\delta \backslash i$	5(T)	6(G)
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$\delta(H), \text{prev is } H$	$4.05 * 10^{-4} * 0.5 * 0.2$ $= 4.05 * 10^{-5}$	←	$4.05 * 10^{-5} * 0.5 * 0.3$ $= 6.075 * 10^{-6}$	←
$\delta(H), \text{prev is } L$	$2.7 * 10^{-4} * 0.4 * 0.2$ $= 2.16 * 10^{-5}$		$6.075 * 10^{-5} * 0.4 * 0.3$ $= 7.29 * 10^{-6}$	
$\delta(L), \text{prev is } H$	$4.05 * 10^{-4} * 0.5 * 0.3$ $= 6.075 * 10^{-5}$	↑	$4.05 * 10^{-5} * 0.5 * 0.2 =$ $7.29 * 10^{-6}$	
$\delta(L), \text{prev is } L$	$2.7 * 10^{-4} * 0.6 * 0.3$ $= 4.86 * 10^{-5}$		$6.075 * 10^{-5} * 0.6 * 0.2$ $= 7.29 * 10^{-6}$	←

$\delta \backslash i$	7(C)		8(A)	
$\delta(H), \text{prev}$	$7.29 * 10^{-6} * 0.5 * 0.3$ $= 1.0935 * 10^{-6}$	←	$1.0935 * 10^{-6} * 0.5 * 0.2$ $= 1.0935 * 10^{-7}$	
$\delta(H), \text{prev}$	$7.29 * 10^{-6} * 0.4 * 0.3$ $= 8.748 * 10^{-7}$		$8.748 * 10^{-7} * 0.4 * 0.2$ $= 6.9984 * 10^{-8}$	
$\delta(L), \text{prev}$	$7.29 * 10^{-6} * 0.5 * 0.2$ $= 7.29 * 10^{-7}$		$1.0935 * 10^{-6} * 0.5 * 0.3$ $= 1.64025 * 10^{-7}$	
$\delta(L), \text{prev}$	$7.29 * 10^{-6} * 0.6 * 0.2$ $= 8.748 * 10^{-7}$	←	$8.748 * 10^{-7} * 0.6 * 0.3$ $= 1.57464 * 10^{-7}$	

*So overall, we got that the maximal probability for this sequence is $1.64025 * 10^{-7}$*

Question 2

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12:03

Input: An integer n , parameters $q(w|t, u, v)$ and $e(x|s)$.

Definitions:

- Define $\mathcal{K}$ to be the set of possible tags.
- Define $\mathcal{K}_{-2} = \mathcal{K}_{-1} = \mathcal{K}_0 = \{*\}$, and $\mathcal{K}_k = \mathcal{K}$ for $k \in [n]$
- Define V to be the set of possible words.

Filling the rest of the algorithm:

Initialization: $\pi(0, *, *, *) = 1$

Algorithm: $\forall i \in [n], \forall u \in \mathcal{K}_{i-2}, \forall v \in \mathcal{K}_{i-1}, \forall z \in \mathcal{K}_i$

we calculate $\pi(i, u, v, z)$

$$= \max_{t \in \mathcal{K}_{i-3}} (\pi(i-1, t, u, v) \times q(z|t, u, v) \times \max_{x \in V} e(x|z))$$

Return: $\max_{u \in \mathcal{K}_{n-2}, v \in \mathcal{K}_{n-1}, z \in \mathcal{K}_n} \pi(n, u, v, z) \times q(STOP|u, v, z)$

Besides extending the sliding window size to 3, we also do not know in advance the current word, so we need to maximize over the emission:

$$\max_{x \in V} e(x|z)$$